SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_fourgee_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	fourgee
Selected Domain	Domain 1: The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 4: Quality Education (main). Related: SDG 8 and SDG 10
Project / Solution Name	Oroboros: Consume. Regrow. Plan.
Name (Leader)	KHAIRUL ZARIFF ADEEB BIN KHAIRUL ZIAD
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University Affiliation(s)	Universiti Teknologi MARA

Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing which tasks inside jobs are done by people faster than training can respond. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025). The Ministry of Science, Technology and Innovation (MOSTI) has said that more than 600,000 workers will need reskilling within three to five years. Both figures are quoted from the Domain 1 brief.

Four groups are affected: working adults whose roles are being redesigned, fresh graduates and career switchers choosing a direction, HR and learning-and-development managers deciding whom to train for what, and TVET and university career centres advising students. A typical case is an insurance claims processor whose employer has just announced an AI triage pilot. She has about three months, six hours a week and no training budget.

Existing guidance fails these people in three ways. Skills frameworks and course lists are static and out of date when published. The question “will AI take my job?” cannot be answered honestly, because AI usually absorbs individual tasks, not whole job titles. And most tools hide their sources and give one confident prediction where the evidence supports only a range.

This matters for decent work and productive employment (SDG 8), for relevant skills (SDG 4) and for fairness (SDG 10), because the workers with the least access to training are the least able to adapt.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Oroboros is a free web app that treats a job as a bundle of tasks. The user picks one of 923 O*NET occupations (or a synthetic persona), ticks the tasks they do and the hours each takes, and sets how many hours a week they can learn.

The engine uses published AI-exposure research and public activity signals rather than running a generative model. Exposure estimates and scenario assumptions help users explore task-level change; they are not predictions of job loss. The documented momentum layer compares recent activity with historical activity and caps the multiplier between 0.8 and 1.2. Qwen2.5-7B-Instruct appears in the optional integration design but is not connected in the live demo. The team did not use RunPod or a GPU workload.

The app computes absorption $A = \min(1, E \times m \times s)$ under three scenarios (slow 0.6, gradual 1.0, shockwave 1.6; stated assumptions), ranks real occupations by Fit × Resilience, suggests a blend of two real roles, and outputs an ordered plan with free courses, hours and a proof task. Every task and role carries SDG tags, and a purpose slider re-ranks roles by SDG weight.

The team confirms completion of the mandatory GenAI, Database and Cloud Deployment workshops. Relevant skills to apply include structured data pipelines, transparent AI limitations and web deployment. Replit was used for documentation, the timed video script and the completed nine-slide pitch deck.

The tool supports a person’s own decision. It never rates people and never makes hiring decisions.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Front end: static HTML, CSS and vanilla JavaScript hosted on GitHub Pages; browser-based scoring. No frontend framework is listed in package.json. Fonts: Syne, Figtree and JetBrains Mono via Google Fonts.

Data: O*NET 31.0, published exposure datasets, curated mappings and static JSON exports. Pipeline dependencies declared in the repository: DuckDB >=1.1, HTTPX >=0.27 and openpyxl >=3.1. pandas is not listed in the current pipeline manifest.

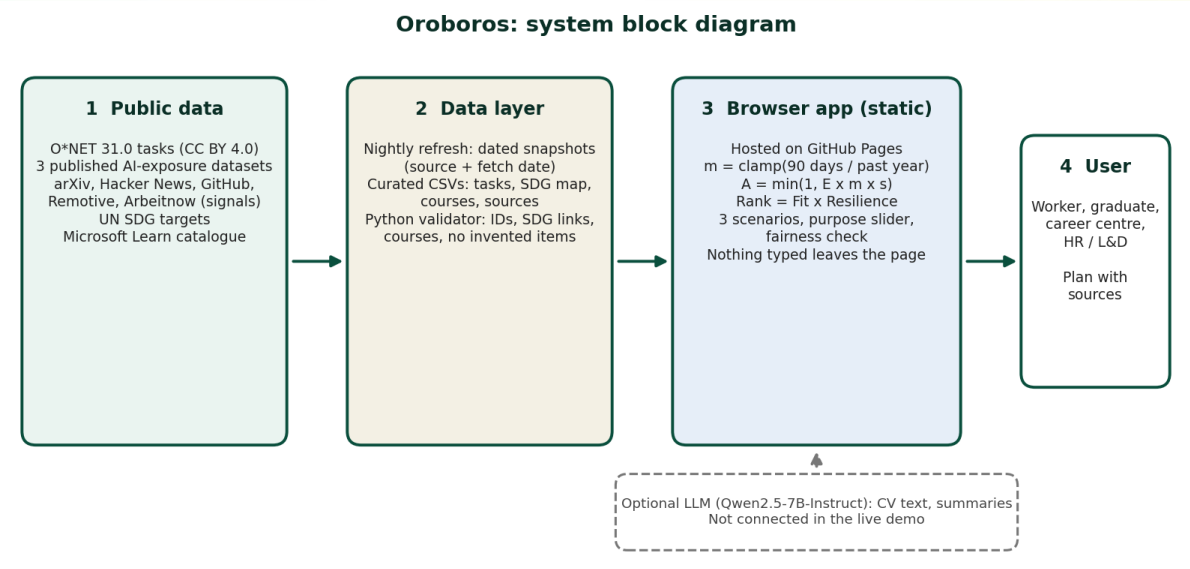
Public-source pipeline: arXiv, Hacker News (Algolia), GitHub REST, Remotive and Arbeitnow; Microsoft Learn Catalog and UN Stats SDG. The repository describes dated snapshots and fallback behaviour; successful refresh runs have not been verified in this review.

Optional AI model: Qwen2.5-7B-Instruct (open weights), not connected in the live demo.

Hosting: the team fork is live on GitHub Pages. Three recent Pages deployment runs succeeded. Nightly refresh is configured but successful scheduled runs on this fork are not yet verified.

Development tools: Claude Code (exact version unconfirmed) for development assistance; Higgsfield AI for web-app design; Replit Agent for documentation and video-script preparation. Replit was used to create and export the completed nine-slide pitch deck.

Architecture: public data sources → Python/DuckDB pipeline → static JSON → GitHub Pages → browser-based scenario scoring → roles and learning plan. Replit supports documentation, video-script and completed pitch-deck work; it is not the app host. RunPod was not used because the project has no GPU workload. The team reports that the administration confirmed RunPod is not required for this project. No sponsor credits are claimed.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design starts with a synthetic claims processor facing an AI triage pilot, limited time and no training budget. We reframe the problem from predicting job disappearance to exploring task-level change and practical learning options. The prototype follows five steps: job intake, capability radar, task exposure, adjacent occupations and a learning plan. Public-source references, explicit scenario assumptions and uncertainty ranges aim to make recommendations understandable. The repository includes engine and server test code; execution and results have not been verified in this document review. Real-user research is not yet established. Proposed Phase 2 validation: observe at least ten workers, students or career-centre staff; measure time to plan, source coverage and whether participants identify an actionable next course. Priorities include Malaysian-context validation, accessibility, language support and evidence-backed sponsor integration. These are proposed activities, not completed research or measured outcomes.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Also Targets 4.3, 8.2 and 8.5 (named in the Domain 1 brief). Source: UN SDG targets list, sdgs.un.org.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	56% of Malaysian professionals use AI at work, but only 26% have formal training (Hays Asia Salary Guide 2025). More than 600,000 workers need reskilling within 3–5 years (MOSTI). Both figures are cited as quoted in the Domain 1 brief; we have not re-checked the original publications. We have no measured user baseline yet; Phase 2 user tests will set it.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	(1) Time from opening the tool to a complete plan, in minutes. (2) Share of recommendations (roles, tasks, courses, figures) that carry a source and a date. (3) Share of test users who name one concrete next skill and one free course.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Targets to be measured in Phase 2 (not yet results): 100% of recommendations carry a source and date; at least 80% of ten or more pilot users reach a plan in five minutes or less; at least 70% name a concrete next course. Proposed evidence: observed test sessions and automated source/data checks. These are planned validation methods, not completed results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

Workshop completion confirmed by the team. Leader review/signature and third-party licensing clearance remain required before submission.

- ☐ ✓ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops (team-confirmed).
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. The team confirms that no work was created before the hackathon.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET 31.0 Database (USDOL/ETA)	dataset	CC BY 4.0
GPTs are GPTs exposure labels (OpenAI)	dataset	MIT; upstream LICENSE checked. Copyright (c) 2024 OpenAI. Include its full notice with redistributed material.

Anthropic Economic Index	dataset	CC-BY for data; MIT for code, per upstream dataset README. CC-BY version is not specified there; metadata still says MIT. Preserve attribution and verify release-specific terms.
Felten et al. AIOE	dataset	No explicit license grant found upstream. Authors request citation, but redistribution permission remains unresolved; attribution alone is not permission.
arXiv, Hacker News (Algolia), GitHub, Remotive, Arbeitnow, Microsoft Learn, UN Stats SDG APIs	public data services	Each provider's API terms
Qwen2.5-7B-Instruct (optional, not connected)	model	Apache-2.0 as declared in repository LICENSE_NOTICES.md.
Python 3; DuckDB, HTTPX and openpyxl (documented pipeline dependencies).	code	PSF License; MIT; BSD-3-Clause; MIT respectively.
Syne, Figtree, JetBrains Mono (Google Fonts); DuckDB, HTTPX, openpyxl; optional FastAPI, Uvicorn, Pydantic.	code	Fonts: SIL Open Font License. DuckDB/openpyxl/FastAPI/Pydantic: MIT; HTTPX/Uvicorn: BSD-3-Clause. Verify transitive dependency notices before signing.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. The team confirms that no employment, internship or consulting agreement applies under which an employer could claim ownership of the hackathon work.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25 October 2026